

ET4437: Distributed Computing and Java

Group Project

Given: 5/11/2024

Type of assignment: group project; groups of two-three students

Group demonstration: week 13

Brightspace submission due: 6/12/2023

Submission content: PDF report and ZIP project file

Worth: 30% (of total grade)

Specification and Deliverables:

Each group must build a Java Application that implements the popular game **Tic Tac Toe (3 x 3 board)**. The Java application must use a SOAP Web Service that will be provided. The Web Service will access data stored on an MySQL server.

The requirements for the Java Application are as follows:

1. The application has to be built as a Java GUI using the Swing library. You will need to create a few windows that may show up on the screen at the same time.
2. First, a small window must load when the application start, offering two choices: create an account (register) or login using existing credentials.
3. Build a login window to enter credentials. This could be a separate Java file.
4. Build a registration window to create a new account. This could be a separate Java file.
5. Once logged/registered, the original window disappears, and the main game window must pop up on the screen. The window must visualize the following components:
 - a. A button to access a score system that will keep track of the number of wins, losses and draws the player has. Note this should open a new window (a new Java class file).
 - b. A button to access a leader board, showing the statistics (wins, losses and draws) for each registered user. Note this should open a new window (a new Java class file).

- c. A button to create a new game. Once game is created, the user must make the first move and wait for someone to join the game.
 - d. A table showing a list of open games that you can join (see JTable).
 - e. You must be able to join a game by selecting it from the list of open games. When the user joins the game, the 3x3 game board will show up, and the user must make the next move.
 - f. Delete a game automatically (a game you've created) if it hasn't started (no one joined the game) within 15mins of creating it.
6. Threading is essential in this project. Your game has two players at remote locations – without threads you will not be able to control who can take a turn at a time. Use can use either **MyThread extends Thread** implementation or **MyThread implements Runnable**. The latter is preferable.
 7. You must use the Web Services provided by **TicTacToeWS** for login, registration, playing the game and accessing the database.
 8. For the development of the Web client application Java you will need to import **tictactoe.sql** database (using phpMyAdmin) and the Web Service **TicTacToeWS.zip** (in NetBeans). To run your Web service client application, start the MySQL server in XAMPP and deploy/run **TicTacToeWS** in NetBeans.
 9. It is important that you make no changes to **TicTacToeWS** - only deploy it (run) or undeploy it (stop). When you run the web service project, the Javadoc tool is also executed and produces HTML documentation for the source files. The documentation can be accessed from the project directory: *TicTacToeWS/target/apidocs/index.html*. The generated documentation is more easier to read than the java files of the Web Service.
 10. For the demonstration, both the MySQL server and the TicTacToe Web Service will be running on another machine, and you will need to update the provided WSDL URL (that includes the IP address of the remote machine) required for your application. Information on how to do that will be provided later on. You only need to load your Web Client application in NetBeans. Before the demonstration, make sure to test your java application after updating the WSDL link.
 11. During the demonstration, you'll need to showcase the following features of your application: registering a new user, login as the user, accessing the score board and leader board, creating a new game, joining an existing game, playing a game with you project partner.
 12. You will also need to write a group report on the design of the final system. The report must include: the project specification, the structure of your application, list of the source Java files and what they do, how the workload was split between you and your project partner, and the project deliverables you accomplished. Make sure you include screenshots of your java application. The report must be submitted on Brightspace as a PDF document, not larger than 20 pages.

13. In addition to the PDF report, your submission must also include the Maven project (your client application) as an exported ZIP file. It's important to include comments in your java files.

Grading:

This project is worth **30%** of the total module grade. The grade is based on:

- Demonstration of the Java application in the lab. All group members must be present.
- Well written report.
- All features implemented and working, e.g. login, registration, joining a game, etc.

Files that will be provided to you:

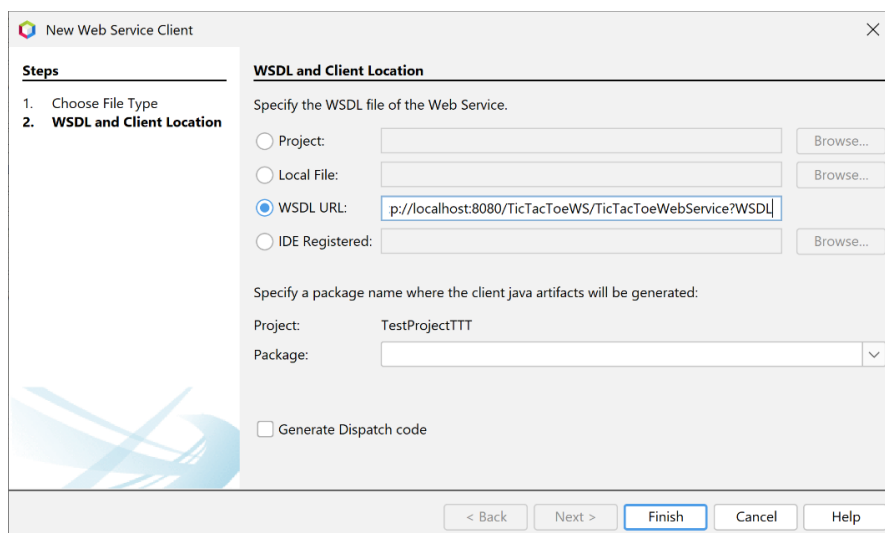
- **TicTacToeWS** – a Netbeans project that implements the Tic Tac Toe Web Service. You will need to use this Web Service for building and testing your game application.
- **tictactoe.sql** – an MySQL database. You'll need to import the database **tictactoe**. This database will be required for building and testing your game application.

Required Software for Development:

- **Netbeans** – with GlassFish 7.0.6 (or newer) and JDK 20.
- **XAMPP** –needed for Apache Web server and MySQL.

Creating a Web Client Application in Netbeans:

1. Create a new project in Netbeans: File-> New Project-> Java with Maven->Java Application
2. Right click on the new project and select New->Web Service Client ...
3. Enter the Location of the WSDL file as shown below, and click Finish.



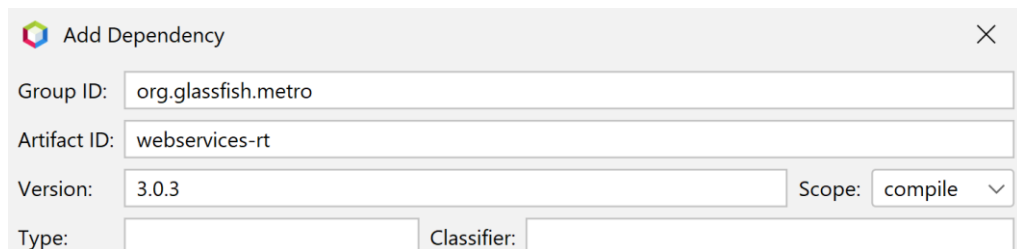
4. In pom.xml, replace

```
<groupId>org.jvnet.jax-ws-commons</groupId>  
<artifactId>jaxws-maven-plugin</artifactId>  
<version>2.3</version>
```

with

```
<groupId>com.sun.xml.ws</groupId>  
<artifactId>jaxws-maven-plugin</artifactId>  
<version>4.0.1</version>
```

5. Add the following dependency (see Lab 6):



Add Dependency

Group ID:

Artifact ID:

Version: Scope:

Type: Classifier:

6. Add Java files to your project and make sure you import the two class files associated with the web service:

```
import com.tttws.TicTacToeWS;  
import com.tttws.TicTacToeWebService;
```

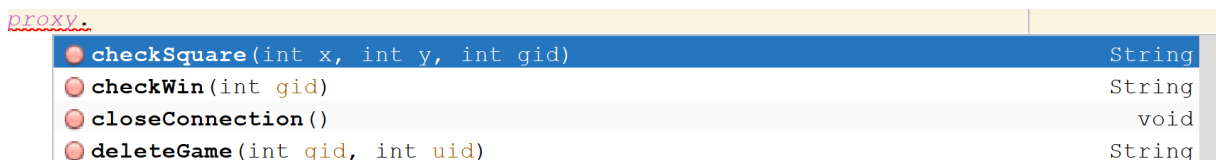
7. Also, two static class variables must be declared:

```
private static TicTacToeWebService service;  
private static TicTacToeWS proxy;
```

and initialized in the constructor:

```
service = new TicTacToeWebService();  
proxy = service.getTicTacToeWSPort();
```

8. Any of the operations offered by the web service can be accessed through proxy:



```
proxy.  
• checkSquare(int x, int y, int gid) String  
• checkWin(int gid) String  
• closeConnection() void  
• deleteGame(int gid, int uid) String
```

Should you have any questions, please contact me (petar.iordanov@ul.ie).

PS. Notes on how to write a good technical report can be found at

<https://www.sussex.ac.uk/ei/internal/forstudents/engineeringdesign/studyguides/techreportwriting>